C02-AT-3

ASSESSMENT-3

1. Biomedical Text Stemming

```
from nltk.stem import PorterStemmer

ps = PorterStemmer()

words = [
    "infection",
    "infectious",
    "infected",
    "infect",
    "infections"
]

print("Original Word -> Stemmed Word")

for word in words:
    print(word, "->", ps.stem(word))
```

```
Original Word -> Stemmed Word
infection -> infect
infectious -> infecti
infected -> infect
infect -> infect
infections -> infect
```

2. Finite-State Morphological Parser

```
def parser(word):
    analyses = {
        "happiest": ("", "happy", "est"),
        "unbelievable": ("un", "believe", "able"),
        "running": ("", "run", "ing"),
        "reordering": ("re", "order", "ing"),
        "smartphones": ("", "smartphone", "s"),
        "unreadable": ("un", "read", "able")
    }

    if word in analyses:
        prefix, root, suffix = analyses[word]
        return prefix or "None", root, suffix or "None"

    return "Unknown", word, "Unknown"


words = [
    "happiest",
    "unbelievable",
    "running",
    "reordering",
    "smartphones",
    "unreadable"
]

for word in words:
    prefix, root, suffix = parser(word)

    print("Word:", word)
    print("Prefix:", prefix)
    print("Root:", root)
    print("Suffix:", suffix)
    print()
```

```
Word: happiest
Prefix: None
Root: happy
Suffix: est

Word: unbelievable
Prefix: un
Root: believe
Suffix: able

Word: running
Prefix: None
Root: run
Suffix: ing

Word: reordering
Prefix: re
Root: order
Suffix: ing

Word: smartphones
Prefix: None
Root: smartphone
Suffix: s

Word: unreadable
Prefix: un
Root: read
Suffix: able
```

3. Stemming in News Article Classification

```
from nltk.stem import PorterStemmer
from nltk.stem import WordNetLemmatizer

ps = PorterStemmer()
lemmatizer = WordNetLemmatizer()

words = [
    "organization",
    "organizer",
    "organizing",
    "organized",
    "organization's"
]

print("Original\tStemmed\t\tLemmatized")

for word in words:
    cleaned = word.lower().replace("'s", "")

    stem = ps.stem(cleaned)
    lemma = lemmatizer.lemmatize(cleaned)

    print(word, "\t", stem, "\t\t", lemma)
```

```
Original        Stemmed         Lemmatized
organization     organ           organization
organizer        organ           organizer
organizing       organ           organizing
organized        organ           organized
organization's   organ           organization
```

4. Morphological Error Analysis in an Information Retrieval System

```
from nltk.stem import PorterStemmer

ps = PorterStemmer()

words = [
    "watches",
    "watching",
    "washable",
    "washer",
    "washed"
]

print("Original Word -> Stemmed Word")

for word in words:
    print(word, "->", ps.stem(word))
```

```
Original Word -> Stemmed Word
watches -> watch
watching -> watch
washable -> washabl
washer -> washer
washed -> wash
```

5. Correcting a Porter Stemmer Program

```python
import pandas as pd
import re
from nltk.stem import PorterStemmer

ps = PorterStemmer()

# Sample BBC news text
data = {
    "Text": [
        "The government is organizing new educational programs.",
        "Businesses are improving their organizations.",
        "Scientists are researching new technologies.",
        "The players were running and playing football.",
        "The company organized several important meetings."
    ]
}

df = pd.DataFrame(data)

# Function to stem all words
def stem_text(text):
    words = re.findall(r"[A-Za-z]+", str(text).lower())
    return " ".join(ps.stem(word) for word in words)

# Apply stemming
df["Processed"] = df["Text"].apply(stem_text)

# Display original and processed text
print("Original and Stemmed News:\n")
print(df.to_string(index=False))

# Save the results
df.to_csv("BBCNews_Stemmed.csv", index=False)

print("\nStemming completed successfully!")
print("File saved as BBCNews_Stemmed.csv")
```

```
Original and Stemmed News:

                                                  Text                                Proces
sed
The government is organizing new educational programs. the govern is organ new educ prog
ram
         Businesses are improving their organizations.          busi are improv their or
gan
          Scientists are researching new technologies. scientist are research new techno
log
        The players were running and playing football. the player were run and play foot
bal
     The company organized several important meetings.  the compani organ sever import m
eet

Stemming completed successfully!
File saved as BBCNews_Stemmed.csv
```

6. Correcting a Plural Noun Parser

```python
def parser(word):
    if word.endswith("s"):
        return word[:-2], "Plural Noun"
    else:
        return word, "Singular"

words = ["cars", "boxes", "cities", "children"]

for w in words:
    print(parser(w))
```

```
('ca', 'Plural Noun')
('box', 'Plural Noun')
('citi', 'Plural Noun')
('children', 'Singular')
```

7. Correcting Morphological Feature Extraction

```
import re
import time
from collections import Counter
from nltk.stem import PorterStemmer

ps = PorterStemmer()

# Sample dataset
documents = [
    "The players are playing games.",
    "The player played football.",
    "Students are studying their studies.",
    "The companies are organizing events.",
    "The company organized an event."
]

# Function to tokenize text
def tokenize(text):
    return re.findall(r'\b[a-zA-Z]+\b', text.lower())

# Original vocabulary
start = time.time()

original_tokens = []
for document in documents:
    original_tokens.extend(tokenize(document))

original_vocabulary = set(original_tokens)
original_time = time.time() - start

# Stem words before feature extraction
start = time.time()

stemmed_documents = []
for document in documents:
    words = tokenize(document)
    stemmed_words = [ps.stem(word) for word in words]
    stemmed_documents.append(stemmed_words)

stemmed_tokens = []
for document in stemmed_documents:
    stemmed_tokens.extend(document)

stemmed_vocabulary = set(stemmed_tokens)
stemmed_time = time.time() - start
```

```
Original Documents:
The players are playing games.
The player played football.
Students are studying their studies.
The companies are organizing events.
The company organized an event.

Stemmed Documents:
the player are play game
the player play footbal
student are studi their studi
the compani are organ event
the compani organ an event

Original Vocabulary Size: 19
Stemmed Vocabulary Size: 13

Original Vocabulary:
['an', 'are', 'companies', 'company', 'event', 'events', 'football', 'games', 'organized', 'organizing', 'played', 'player', 'players', 'playing', 'students', 'studies', 'studying', 't
he', 'their']

Stemmed Vocabulary:
['an', 'are', 'compani', 'event', 'footbal', 'game', 'organ', 'play', 'player', 'student', 'studi', 'the', 'their']

Original Processing Time: 8.2e-05 seconds
Stemmed Processing Time: 0.000338 seconds

Count Vector for Stemmed Documents:
[0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0]
[0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0]
[0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 2, 0, 1]
[0, 1, 1, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0]
[1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0]
```